

Optics
21-22

1. A silica optical fiber has a core refractive index n_1 of 1.50 and a cladding refractive index (n_2) of 1.47.

g) Calculate the **Critical Angle** (Φ_c) at the core-cladding interface.

h) Calculate the **Numerical Aperture (NA)** of the fiber.

i) Determine the **Acceptance Angle** (θ_a) for the fiber in air ($n_0 = 1$).

2. Using Snell's Law, derive the relationship: $NA = \sqrt{n_1^2 - n_2^2}$. Show how this relates to the relative refractive index difference (Δ).

3. An optical fiber in air has an NA of 0.4. If a specific skew ray changes direction by 100° at each reflection ($\gamma = 50^\circ$), calculate its acceptance angle. Compare this with the acceptance angle for meridional rays.

3. What is Monte Carlo Method? Write down the difference between monte-carlo and ...

Patuakhali Science and Technology University

Department of Computer Science and Information Technology

6th Semester (Level-3, Semester-II), Midterm Examination of B.Sc. Engg.(CSE), July-Decem./2024, Session: 2021-22

Course Code: CIT-323 Course Title: Simulation and Modeling

Full Marks: 15 Duration: 1 hour

1. Define system and model with examples. Distinguish between analytical solution and simulation. 03
Demonstrate a flowchart of steps in a sound simulation study.
2. List the steps of simulation for a single-server queueing system. Formulate the problem statement and describe the intuitive explanation of a single-server queueing system. 05
3. What is Monte Carlo Method? Write down the difference between monte-carlo and stochastic simulation. 02
Define Monte-Carlo method. Solve the integrals $I = \int_1^5 \frac{x^4}{3} dx$ by monte-carlo method. [USE Random Number 48, 36, 44, 41, 46, 18, 32, 41, 23, 23 for X coordinate and Random Number .52, .18, .30, .88, .21, .25, .57, .82, .9, .63 for Y coordinate, you have a right to do scaling, if required]

$$2 \times (4 \times 2.10)$$

$$\frac{208}{2}$$

simulation

Patuakhali Science and Technology University

6th Semester (Level-3, Semester-II), Mid Examination of B.Sc. Engg. (CSE), January-June: 2025

Course Code: CCE-21 Course Title: Computer Peripheral and Interfacing

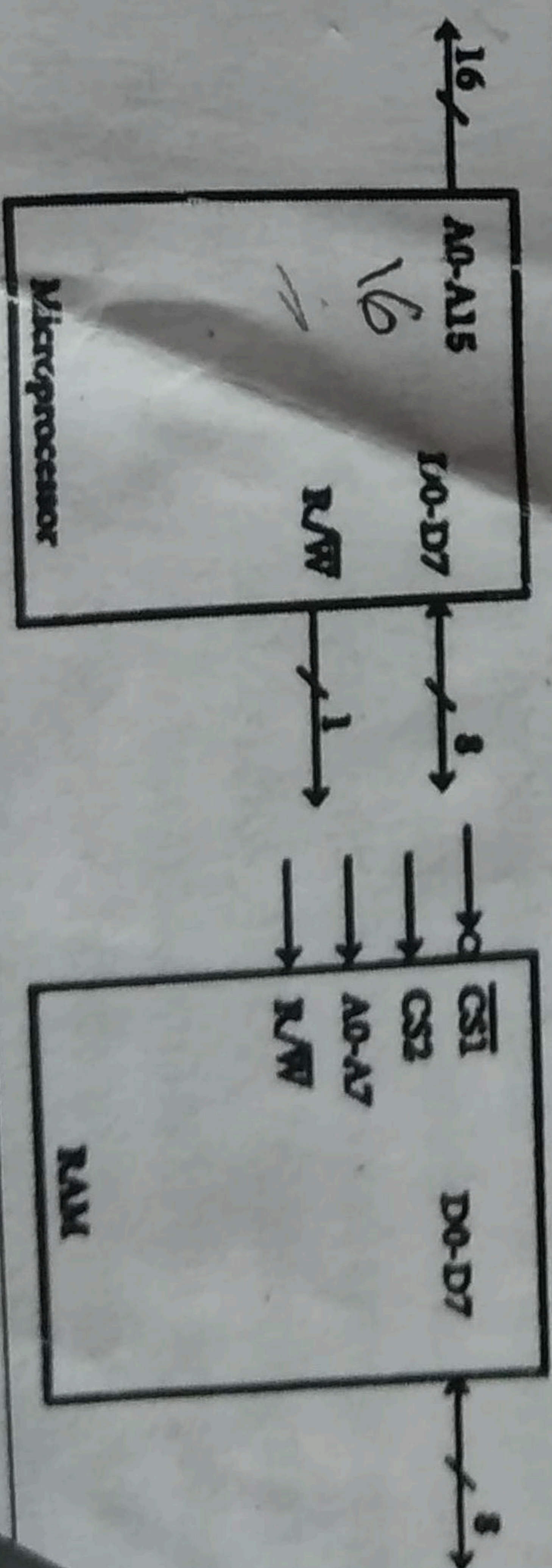
Credit Hour: 3.0 Full Marks: 15 Time : 1 Hour

Fill in the Blanks

e.	The Internet of Things (IoT) is the network of smart _____ objects embedded with sensors, computation units, and network connectivity.	2.5
f.	The primary goal of IoT is to "connect the _____".	
g.	The term "Internet of Things" was likely coined by _____ in 1999. IoT systems enable _____ monitoring of equipment and processes to improve efficiency.	
h.	A major challenge in IoT implementation caused by different protocols and architectures is called _____.	
2)	MCQ in the opposite page	2.5

3)	d. Why can a hacked IoT device become a serious threat to other devices and systems in a network? e. Why does the large number of IoT sensors create challenges for data processing and analytics? f. Why is interoperability important for the successful operation of IoT devices from different manufacturers?	3
4)	Differentiate between a) conditional and unconditional programmed I/O. b) buffer and spooler	04

5)	Assume the following microprocessor and the RAM chip: Draw a neat logic diagram showing connections between the above microprocessor and the RAM chip using only the signals shown to include the memory map 020016. Use linear decoding.	03
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Peripheral

2. Write only ~~A~~ answer key as with text

- i. In a smart home IoT system, multiple sensors such as temperature, smoke, door, and power sensors communicate locally using technologies like Wi-Fi or Bluetooth before sending their data to the Internet through an IoT gateway. If the gateway suddenly fails, which functionality would be most directly affected? A) Data sensing at the device layer B) Local wireless communication among sensors C) Communication between the local IoT network and the Internet D) Data storage inside individual sensors.
- ii. In the traditional IoT data flow architecture, data generated by edge devices is first transmitted through local connectivity to an edge gateway and then forwarded through the backhaul network to the cloud for storage and analysis. If network latency between the gateway and the cloud becomes very high and the system must support real-time decision making, which architectural solution would most effectively reduce response delay? A) Increasing the number of sensors in the network B) Performing data analytics directly at the edge gateway C) Sending all raw data to the cloud before processing D) Replacing wireless sensors with wired sensors.
- iii. In many IoT architectures, sensors communicate with each other or with a gateway using local connectivity technologies such as Bluetooth, Wi-Fi, or WirelessHART instead of directly connecting to the Internet. What is the main design reason for using these local communication technologies in IoT networks? ~~A~~) Sensors cannot generate digital data B) Local protocols reduce power consumption and simplify device communication C) Internet connectivity cannot support sensor networks ~~D~~) Sensors must always communicate through wired connections.
- iv. In a traditional IoT system, sensed data is often transmitted from devices to the cloud where processing and analysis take place before users or administrators access the results. If a system designer moves data processing and decision-making from the cloud to the edge of the network, what is the most important architectural advantage of this change? A) Increased dependence on centralized servers B) Faster response time and reduced bandwidth consumption C) Elimination of IoT gateways D) Removal of the need for sensors.
- v. In a long-range IoT monitoring system using LoRa technology, sensor nodes transmit data to a LoRa gateway which then forwards the information to a cloud server through Internet connectivity. In this architecture, which component primarily acts as the intermediary that converts low-power sensor network communication into Internet-compatible communication? A) Cloud server B) Mobile application C) IoT gateway D) Sensor node.

Patuakhali Science and Technology University

Department of Computer Science Information Technology(CSIT)

Midterm Examination, July-December-2021

Program: B. Sc. Engg.(CSE)

Session : 2018-19

Course Code : CIT-324

Course Title : Simulation and Modeling Sessional

Full Time: 1 Hours

Full Marks: 15

Answer the marked question

1. Calculate the area of an irregular figure by Monte Carlo Method. 7.5
2. Simulate the Gambling Game by Monte Carlo Method and analyze profit-loss. 7.5
3. Find the value of a numerical integration by Monte Carlo Method of any differential equation. 7.5
4. Determination of the value of π by Monte Carlo Method. 7.5
5. Solve Random walk by Monte Carlo Method. 7.5
6. Simulate Reliability Problem. 7.5
7. Implement Kolmogorov test with midsquare random numbers. 7.5
8. Implement Chi-Square test with multiplicative congruential random numbers. 7.5
9. Implement chi-square test for autocorrelation with additive congruential random numbers. 7.5
10. Implement poker test with multiplicative congruential random numbers. 7.5

Patuakhali Science and Technology University

Department of Computer Science Information Technology(CSIT)

Midterm Examination, July-December-2024

Program: B. Sc. Engg.(CSE)

Session : 2021-22

Course Code : CIT-324

Course Title : Simulation and Modeling Sessional

Full Time: 1 Hours

Full Marks: 7.5

< Answer the marked question / >

10 60 75 100

5. A drunkard moved from a point to a destination and takes step in **four** directions, backward, forward, right, and left. The probabilities associated with these are 10%, 50%, 15% and 25% respectively. The distances covered to backward, forward, right, and left are 30 cm, 80 cm, 60 cm and 40 cm respectively. Simulate the walk for 20 steps and find the location at end of 20 steps while starting point is (0, 0) on the X-Y scales. [Use Random Number: 6, 2, 0, 6, 2, 8, 5, 7, 7, 9, 8, 4, 8, 2, 6, 2, 1, 3, 0, 8, 4]
6. A machine center has **four** identical bearings, which fail according to the following probability distribution in table A. The present maintenance policy is to change a bearing as and when it fails. When bearing fails machine center stops, a technician is called to replace the failed bearing with a new one. The time between the failure of the bearing and reporting of the technician (delay time) is random and is distributed as table B.

Table: A	
Bearing Life(Hours)	Probability
1200	0.10
1400	0.25
1600	0.30
1800	0.25
2000	0.10

Table: B	
Delay Time(Minutes)	Probability
2	0.2
3	0.3
4	0.3
8	0.2

It takes 15 minutes to change one bearing, 25 minutes to change two bearing, 30 minutes to change three bearing and 35 minutes to change all three bearings. Downtime of the system costs \$5 per minutes, direct on job costs of the technician is \$30 per hour and the cost of bearing is \$30. The maintenance department is interested in evaluating an alternative policy of replacing all the three bearings, whenever a bearing fails. Simulate the present and proposed policy for 400000 hours and choose the better policy.

simulation mid sessional

Central tendency

→ (5686.125A, 26758.5543, 310.7292)



Patuakhali Science and Technology University (PSTU)

Faculty of Computer Science and Engineering

Department of Computer Science and Information Technology

B.Sc. Engg.(CSE), Level-3, Semester: II, Mid-Term Examination, July-December 2024

Course Code: CIT 321

Full Marks: 15

Course Title: Operating System
Time: 01 Hour

- | | Marks |
|---|-------|
| 1. (a) What are the three main purposes of an operating system? Some computer systems do not provide a privileged mode of operation in hardware. Is it possible to construct a secure operating system for these computer systems? Give arguments both that it is and that it is not possible. | 04 |
| (b) What is the purpose of system calls? The services and functions provided by an operating system can be divided into two main categories. Briefly describe the two categories, and discuss how they differ. | 04 |
| (c) In a smart irrigation system, two processes operate concurrently: <ul style="list-style-type: none">• P_1 (Water Controller) increases the water level count after releasing water into the field.• P_2 (Drain Controller) decreases the water level count when excess water is drained. A shared variable $waterLevel = 10$ is used. Both processes execute simultaneously without synchronization. | 07 |

```
P1() {  
    read(waterLevel);  
    waterLevel = waterLevel + 1;  
    write(waterLevel);  
}
```

```
P2() {  
    read(waterLevel);  
    waterLevel = waterLevel - 1;  
    write(waterLevel);  
}
```

Answer the following questions:

- Analyze how both a race condition and instruction reordering (weak memory model effect) can lead to inconsistent or incorrect waterLevel values.
- Evaluate the possible final values of waterLevel under different execution interleaving and explain why these outcomes violate system correctness.
- Propose and justify a hardware-supported synchronization mechanism (compare_and_swap instruction) to ensure mutual exclusion and briefly explain why software-only solutions may fail on modern CPUs.

OS mid



2102027
sharrafat karim

Patuakhali Science and Technology University (PSTU)
Faculty of Computer Science and Engineering

B.Sc. Engg. 3rd Year 2nd Semester, Mid-Term Examination, January-June 2025

Course Code: CIT 322

Full Marks: 15

N.B.: Figures shown in the right margin indicate full marks

Course Title: Operating System Sessional
Time: 01 Hour

Marks
03 5

1. (a) Use the commands to

- Show home directory of your user
- know what files are in the directory you are in (w)
- go to a directory (w)
- create a folder or a directory (w)
- delete files and directories (w)
- create a file (w)
- To know more about a command and how to use it (w)
- make a file executable and to change the permissions granted to it in Linux
- move files through the command line (w)
- locate a file in a Linux system
- check your connection to a server (w)
- move some data, usually text into a file (w)

(b) Process Control - System Calls

- Linux: Creating processes using fork(), exec(), and wait().
- Windows: Creating processes using the CreateProcess() API in C++.
- Task: Observe "Zombie" processes in Linux and "Background Processes" in Windows Task Manager.

(c) Process Control - System Calls

- Linux: Handling signals (SIGKILL, SIGINT) using signal.h.
- Windows: Process termination and priority manipulation via SetPriorityClass.
- Task: Build a "Mini-Shell" that executes commands and handles Ctrl+C gracefully.

(d) A multi-threaded online ticket booking system maintains a shared variable available_seats = 1. Two users try to book the last remaining seat at the same time. Without proper synchronization, both threads may successfully book the seat, causing overbooking. Implement a C++ program where two threads attempt to book the last seat simultaneously. Use CAS to ensure that only one thread can successfully book the seat. Your program must:

- Prevent race conditions using CAS
- Print which thread successfully booked the seat
- Print a failure message for the other thread

OS

mid sessional

[Figures in the right margin indicate full marks] Answer any 5 of the following questions.

1. (a) Write the real applications of simulation and modeling in different domains. Distinguish analytical solution versus simulation. Demonstrate different ways in which a system might be studied. 04
- (b) State the concept of single-server queueing system. List the steps of simulation for a single-server queueing system. Consider a single-server queueing system, there are several IID random variables interarrival times, service times, and customer delays, and describe the problem statement of the single-server queueing system. 06
- (c) Outline the purpose of inventory system simulation in achieving business objectives. Then, develop a simulation model of an inventory system that incorporates key elements found in a real-world inventory system. 04

2. (a) Demonstrate stochastic process. Why stochastic process is used to simulate the output data? Illustrate correlation function of the process of delays in queue D1, D2,for the M/M/1 queue. 04
- (b) Define verification, validation, and credibility in simulation model. List the issues in an accreditation decision. Illustrate the timing relationship of validation, verification, and establishing credibility. Show the mathematical formulation in a good estimate of the mean of the system using both validation and output analysis. 05
- (c) How to compute variance, covariance, and correlation for simulation study? Explain the inspection approach of statistical procedure for comparing real-world observation and simulation output data. 05

3. (a) Identify the role of randomness for simulation study. List some sources of randomness for common simulation applications. Write the properties of location, scale, and shape parameters of continuous distributions. 04
- (b) Linear Congruential Generators (LCGs) are widely used for pseudo-random number generation. Explain how LCGs produce uniform random numbers in [0,1]. Then describe the inverse transform method and apply it to generate random variates from an exponential distribution with mean β . 05
- (c) State the relevant information (application, density function, parameters, and density curve) for the simulation modeling application of the normal continuous distribution. State the necessity of statistical analysis with simulation output data. Describe and illustrate the transient and steady-state behaviors of a stochastic process. 05

4. (a) Define Computer Simulation and explain its significance in modern engineering and scientific research. Discuss the typical steps involved in the simulation process, from problem formulation to model validation and verification. 02
- (b) Explain the core components of a Discrete-Event Simulation, including entities, attributes, events, the event notice, and the simulation clock. Describe how the Next-Event Time Advance mechanism works to move the simulation forward. 02
- (c) In a pure pursuit, there is a target bomber which moves along a predetermine path (given on table below) and there is a pursuer aircraft who follows the target bombers, redirecting itself towards the target bombers at fixed intervals of time. 10

Target bombers moving path:

Time, t	0	1	2	3	4	5	6	7	8
$xb(t)$	160	161	167	172	181	189	199	215	239
$yb(t)$	0	4	9	11	14	24	34	41	52

Time, t	9	10	11	12	13	14	15		
$xb(t)$	245	254	265	271	289	293	300		
$yb(t)$	45	41	36	39	34	28	12		

The fighter aircraft following the enemy bombers with speed at 30KM/min to destroy it. The fighter aircraft is at position x_f, y_f (20, 30) when it sights the bomber that is at time $t=0$, the time at which the pursuit begins. The fighter corrects its direction after a fixed interval of one minute, so as to point towards a bomber and shoots the bomber by firing a missile as soon as it is within a distance of 10 KM. The pursuit ends, in case the bomber is not shot within 15 minutes of the pursuit, the pursuit is abandoned. Simulate the problem to determine the pursuit is end/abandoned.

Simulation

5. (a) Discuss the factors that affect the accuracy of a continuous simulation. How do the choice of Step Size Δt and the order of the integration algorithm impact the trade-off between computational speed and result precision? 02
- (b) In the context of a Monte Carlo simulation for a gambling game based on a fair coin toss where the player wins \$1 for Heads and loses \$1 for Tails, explain how the Law of Large Numbers influences the simulation's results as the number of trials (N) increases. Specifically, discuss the relationship between the simulated average outcome and the theoretical expected value. If the player wins Heads 3 consecutive toss he/she will get \$10 rewards and get penalty of \$10 for 3 consecutive tails. Simulate the game for 30 toss. Collect the first digit column-wise of random numbers given in the table of question 6(b). 04
- (c) There are a number of moving objects, which chase each other in a serial order like A chases B, B chases C, and C chases D etc. Each object moves towards its target unmindful of the fact that it itself is being targeted by some other object. Depending upon the original location, and speed of motion of the object, each object will take its own time to hit its target. As soon as hit occurs, the chase ends. The initial location of the objects A, B, C, and D are (0,0), (0,10), (0,20), (0,30) respectively and their velocities are 30km/hr, 25km/hr, 20 km/hr and 15km/hr respectively. Object D moves towards a fixed at (30,50), while C moves always in a direction pointing towards D, the object B moves always pointing towards C and object A always pointing towards B. It can be assumed that when the distance between two objects less than 0.005 hit is taken to occur. Identify, which object will hit the target at first and in what time? 08

6. (a) i. Why is probability distribution important in analyzing random variables? Discuss with examples from real-life systems. 04
- ii. Given the Linear Congruential Generator: $X_{n+1} = (aX_n + C) \text{ mod } m$, Where $X_0 = 7, a = 5, c = 3, m = 13$. Generate 5 random numbers.
- (b) i. A sequence of 100 random numbers is given below. Use Chi-Square test with $\alpha = 0.01$ to test these numbers are uniformly distributed. 10

21	81	92	23	96	20	68	57	79	84
82	62	12	08	92	83	74	85	60	49
48	37	65	74	22	11	28	10	55	82
72	95	08	85	79	95	86	11	16	52
70	55	50	87	67	51	72	38	29	62
71	12	07	75	56	34	40	67	24	86
18	82	41	29	63	06	84	01	20	06
06	33	14	79	25	65	57	47	74	68
54	35	81	07	88	96	70	85	29	13
12	91	26	57	30	22	90	03	13	31

- ii. Define autocorrelation. Test the autocorrelation of the given numbers (6(b)i) by employing the chi-squared test with 95% confidence level.
- N.B: You can use chi-squared table below as required.

(c)

Table: Chi-square Distribution

d.f.	.995	.99	.975	.95	.9	.1	.05	.025	.01
1	0.00	0.00	0.00	0.00	0.02	2.71	3.84	5.02	6.63
2	0.01	0.02	0.05	0.10	0.21	4.61	5.99	7.38	9.21
3	0.07	0.11	0.22	0.35	0.58	6.25	7.81	9.35	11.34
4	0.21	0.30	0.48	0.71	1.06	7.78	9.49	11.14	13.28
5	0.41	0.55	0.83	1.15	1.61	9.24	11.07	12.83	15.09
6	0.68	0.87	1.24	1.64	2.20	10.64	12.59	14.45	16.81
7	0.99	1.24	1.69	2.17	2.83	12.02	14.07	16.01	18.48
8	1.34	1.65	2.18	2.73	3.49	13.36	15.51	17.53	20.09
9	1.73	2.09	2.70	3.33	4.17	14.68	16.92	19.02	21.67
10	2.16	2.56	3.25	3.94	4.87	15.99	18.31	20.48	23.21
11	2.60	3.05	3.82	4.57	5.58	17.28	19.68	21.92	24.72
12	3.07	3.57	4.40	5.23	6.30	18.55	21.03	23.34	26.22
13	3.57	4.11	5.01	5.89	7.04	19.81	22.36	24.74	27.69
14	4.07	4.66	5.63	6.57	7.79	21.06	23.68	26.12	29.14
15	4.60	5.23	6.26	7.26	8.55	22.31	25.00	27.49	30.58
16	5.14	5.81	6.91	7.96	9.31	23.54	26.30	28.85	32.00
17	5.70	6.41	7.56	8.67	10.09	24.77	27.59	30.19	33.41
18	6.26	7.01	8.23	9.39	10.86	25.99	28.87	31.53	34.81
19	6.84	7.63	8.91	10.12	11.65	27.20	30.14	32.85	36.19
20	7.43	8.26	9.59	10.85	12.44	28.41	31.41	34.17	37.57

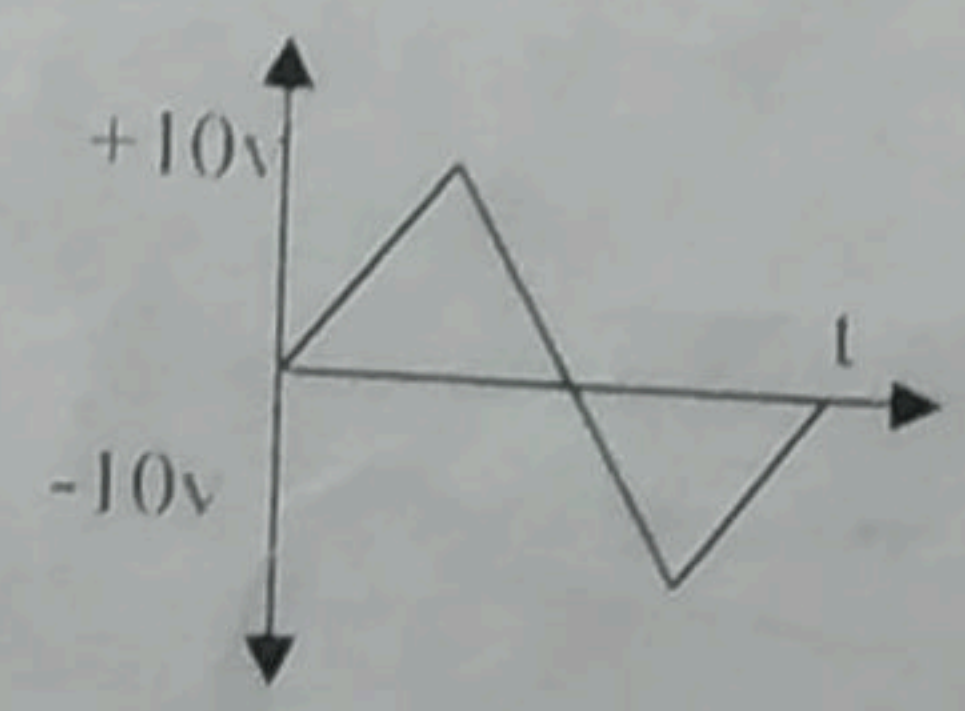
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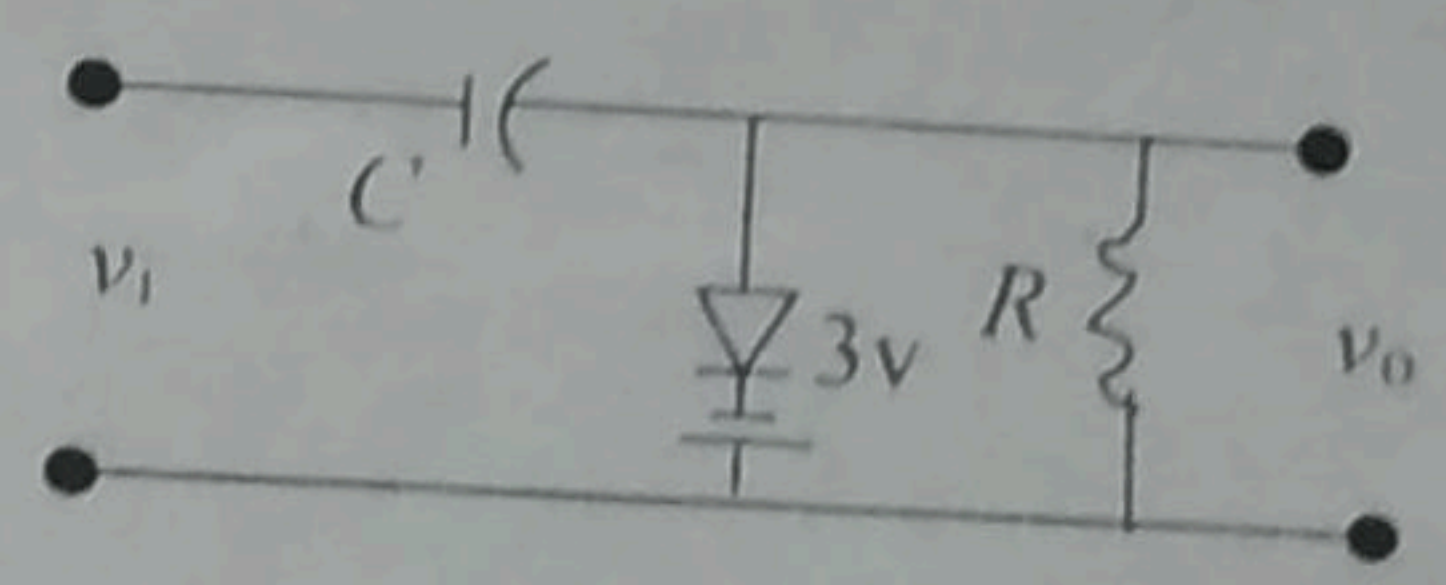
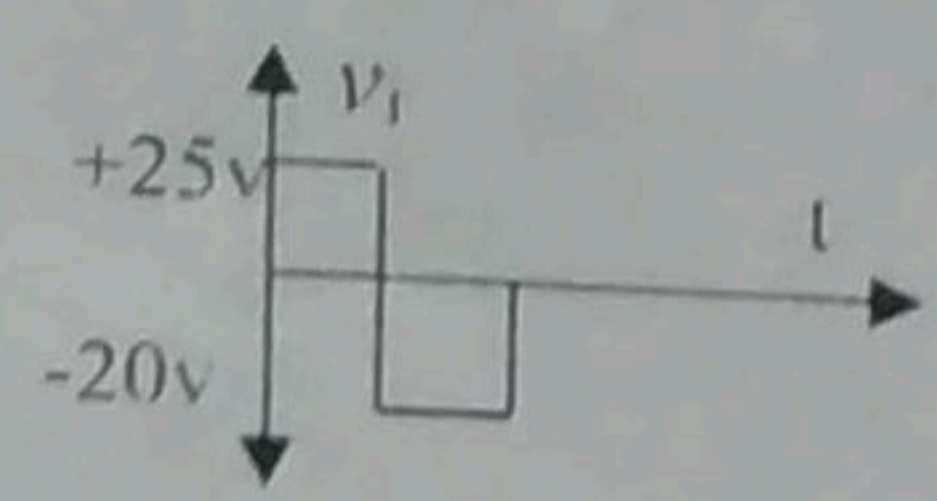
Patuakhali Science and Technology University
6th Semester (L-3, S-II), Final Exam. of B. Sc. Engg. (CSE), Jul-Dec 2024
Course Code: EEE 321 Course Title: Digital Electronics and Pulse Technique
Credit Hour: 3.0 Full Marks: 70 Duration: 3 hours
[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

- Answer any 5 of the following questions
1. (a) How bipolar junction transistor (BJT) works like an electronic switch? Show the configuration of CE transistor switch. 3+2
 - (b) RTL was one of the earliest logic families. Why it is not used much today? 3
 - (c) Design the 3 input DTL NAND gate. Calculate i) noise margin ii) fan out iii) average power considering standard parameters of diode and transistor and $h_{FE} = 35$. 6
 2. (a) Show the structure of TTL NAND gate. Why TTL became dominant in logic families? 3+2
 - (b) Design a weighted-resistor D/A converter. Find the analog output voltage of a 3-bit D/A converter for all possible inputs. Assume $K=1$ 3+3
 - (c) How quantization error is reduced in A/D converter? 3
 3. (a) Express the ideal characteristics of Op-Amp. Show the differences between open loop gain and close loop gain. 3+3
 - (b) Draw the circuit of an inverting amplifier with input voltage 5V; $R_f = 100K\Omega$; $R_{in} = 10 K\Omega$. Find voltage gain, output voltage and current from the circuit. 5
 - (c) Compare the performance of inverting amplifier and non-inverting amplifier. 3
 4. (a) Derive the voltage gain expression for non-inverting amplifier. 4
 - (b) What is voltage follower? Why it is used in impedance matching? 4
 - (c) Define a weighted adder? Draw a 3-input inverting adder with inputs of 2V, 4V and 5V; $R_f = 20K\Omega$; $R_{in} = 10 K\Omega$. Find out output voltage from the circuit. 6
 5. a) Draw the circuit of integrator and derive the output equation of integrator. 5
 - (b) Why is a capacitor added in practical differentiator? 2
 - (c) Define cut-off frequency. How it is calculated? Design a low pass filter for $R = 10K\Omega$; $C = 6\mu F$. 2+2
 6. a) What are the differences between clipper and clamper? Design a clipper circuit which will clip above 5V and below -4V with given input signal and show the output waveform. +3



$$-v_s + IR + 5 = 0$$
$$IR = v_s - 5 \geq 0$$
$$v_s \geq 5$$

- b) Analyze output voltage for given clamping network where $f = 1500 \text{ Hz}$, $R = 20 \text{ Kilo-Ohm}$, $C = 6 \text{ nF}$?



-3
-48

- (c) What's the importance of time constant in clamping circuit?

EEE

[Figures in the right margin indicate full marks. **Split answering of any question is not recommended.**
Write the full question number e.g. 5(B)(iii) before the answer paragraph]

Answer any 5 of the following questions

- 1 A A silica optical fiber with a core diameter large enough to be considered by ray theory analysis has a core refractive index of 1.57 and a cladding refractive index of 1.50. 7=
Determine: 2+2+3
- the critical angle at the core-cladding interface;
 - the NA for the fiber;
 - the acceptance angle in air for the fiber.
- 1 B What are the differences between the general communication system and optical fiber communication system? Explain optical fiber communication system with appropriate block diagram. 7
- 2 A When the mean optical power launched into a 12 km length of fiber is $125 \mu W$, the mean optical power at the fiber output is $5 \mu W$. 7=2+
Determine: 2+2+1
- the overall signal attenuation or loss in decibels through the fiber assuming there are no connectors or splices;
 - the signal attenuation per kilometer for the fiber.
 - the overall signal attenuation for a 16 km optical link using the same fiber with splices at 1 km intervals, each giving an attenuation of 1 dB;
 - the numerical input/output power ratio in (iii).
- 2 B 7=
- A glass fiber has a material dispersion coefficient (D_{mat}) of $80 \text{ ps}/(\text{nm} \cdot \text{km})$ at an operating wavelength of 850 nm. The light source used is an LED with a spectral width ($\Delta\lambda$) of 40 nm. 4+3
 - Calculate the pulse broadening (ΔT) due to material dispersion for a fiber length of 2 km.
 - Calculate the maximum Bit Rate (R_B) that can be supported by this link, assuming the bit rate is limited to: $R_B \approx 1 / (2\Delta T)$
 - An optical receiver uses a photodiode with a quantum efficiency of 70% at a wavelength of 1310 nm. 7/2
 - Calculate the Responsivity (R) of the photodiode.
 - If the received optical power is $5 \mu W$, calculate the resulting photocurrent (I_p).
- 3 A 7=
- What is optical fiber? Show its typical configuration with necessary diagram.
 - Explain in detail the intrinsic absorption loss mechanisms with necessary illustrations. 4+3

Optical

- 3 B A step index fiber has a core refractive index of 1.5 and a core diameter of $50\text{ }\mu\text{m}$. The fiber is joined with a lateral misalignment between the core axes of $5\text{ }\mu\text{m}$. Estimate the insertion loss at the joint due to the lateral misalignment assuming a uniform distribution of power between all guided modes when: 7=
4+3
- (i) There is a small air gap at the joint;
 - (ii) The joint is considered index matched.
- 4 A i. "The intrinsic loss through a multimode fiber joint is independent of direction of propagation." State whether the given statement is true or false. Draw and explain the different causes of radiative losses in fiber-to-fiber joints. 7=
4+3
- ii. In a Wavelength Division Multiplexing (WDM) network, the stability of the transmitter is critical. Explain why "Wavelength Lockers" are used and how "Precision Wavelength" prevents crosstalk.
- 4 B i. What are Passive Components in Optical Networks? Explain their working principles. 7=
2+2+3
- ii. Explain the function of MUX and DEMUX with a real-life example.
- iii. Why is a Wavelength Converter necessary in optical networks? Explain with example.
- 5 A i. Distinguish between Selectivity and Sensitivity in the context of a WDM optical receiver. What happens to the system performance if the receiver has high sensitivity but poor selectivity? 7=
4+3
- ii. A WDM optical link consists of 16 channels. Each channel enters an Erbium-Doped Fiber Amplifier (EDFA) with an input power of -18 dBm . If the EDFA provides a gain of 22 dB :
- a) Calculate the output power of a single channel in dBm .
 - b) Calculate the total aggregate output power of the EDFA in milliwatts (mW).
- 5 B i. A four-port multimode fiber FBT coupler has an input optical power of $60\text{ }\mu\text{W}$ at port 1. The output powers at ports 2, 3, and 4 are measured as $0.004\text{ }\mu\text{W}$, $26.0\text{ }\mu\text{W}$, and $27.5\text{ }\mu\text{W}$ respectively. Calculate the following parameters for the coupler: 7=
3+2+2
- a. Excess loss
 - b. Insertion loss between input and each output port
 - c. Crosstalk
 - d. Power splitting ratio
- ii. Define fusion splicing of optical fibers. Discuss the advantages and disadvantages of this fiber jointing method.
- iii. Explain the mechanism of optical feedback in a laser that leads to oscillation and amplification. Also, describe how this mechanism results in a distinct spectral output from the laser.
- 6 A Discuss the design aspects of optical fiber cables, focusing on: 7=
4+3
- a. the purpose and techniques of fiber buffering;
 - b. the role of strength and structural components in the cable.
- 6 B i. Explain how single-frequency operation is achieved in a Distributed Feedback (DFB) laser, and compare it with the operation of a Fabry-Pérot laser. 7=
4+3
- ii. Describe two widely used fabrication techniques for three-port and four-port optical fiber couplers.

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Dept. of Computer and Communication Engineering
Faculty of Computer Science and Engineering
Patuakhali Science and Technology University
Dumki, Patuakhali-8602, Bangladesh

Final Examination of B. Sc. Engineering in CSE Level: III Semester: II Session: 2021-2022

Course Code

Course Title

July-December 2024

Credit: 03

CCE-321

Computer Peripheral and Interfacing

Time: 03 Hr

Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

- 1 [A.] i. What is the Internet of Things (IoT)? Explain its goal and how it differs from traditional computer networks. 2x7
=14
- ii. List and briefly explain any three major benefits of IoT as mentioned in the slides.
- iii. What is an IoT gateway? Describe its role in providing bidirectional communication and security.
- iv. Name any three real-world IoT applications. Briefly explain how IoT is used in each.
- v. Draw or describe the commonly used 4-layer IoT architecture. Explain the function of each layer. ✓
- 0.5 vi. Differentiate between licensed and unlicensed frequency bands in IoT. Give one example of each.
- vii. What is edge computing in IoT? How does it differ from traditional cloud-based data processing?
- 2 [A.] i. What are the six basic communication criteria used to describe IoT access technologies? Explain each briefly. 2x7
=14
- ii. Differentiate between short-range, medium-range, and long-range IoT access technologies with one example each.
- iii. What are the three classes of constrained nodes defined by RFC 7228? State their RAM, flash, and IP stack capabilities.
- iv. List any five common security attacks in IoT. Explain any two of them briefly.
- v. What is a smart object in IoT? Describe its five essential characteristics as mentioned in the slides.
- vi. Draw and explain the pin diagram of Arduino UNO. Specify the functions of at least five pins.
- vii. What are the two mandatory functions in an Arduino sketch? Explain their purposes with suitable examples.
- 3 [A.] i. Draw a chip select circuit to have the address of an I/O mapped I/O port as E7H. 5
- ii. Draw a chip select circuit to have the address of a memory mapped I/O port as 7EE7H.
- [B.] Assuming signed numbers, find the sign, carry, and overflow flags of: 4
- i. $09_{16} + 17_{16}$
- ii. $71_{16} + A9_{16}$
- iii. $6E_{16} + 3A_{16}$
- iv. $7E_{16} + 7B_{16}$
- [C.] In USB communication, a single transaction involves the transmission of up to three packets. Mention the names of these packets and describe the specific services each one provides during the communication process. 3
- [D.] What is meant by foldback in linear decoding? 2
- Token data, handshake

Peripheral

- 4 [A.] Show the pin configuration of 8255 PPI and separately describe the operation performed by each unit. 5
- [B.] When you press a key on your computer, you are activating a switch. There are many different ways of making these switches. Describe any four of them. 4
- [C.] i. How many address and data lines are required for a 1M x 16 memory chip? 3
 ii. What is the size of a decoder with one chip enable (CE) to obtain a 64K x 32 memory from 4K x 8 chips? Where are the inputs and outputs of the decoder connected? 3
- [D.] Write down the advantages and disadvantages of PDP over CRT and LCD. 2
- 5 [A.] i. If we want to cut a 512 x 512 sub-image out from the center of an 800 x 600 image, what are the coordinates of the pixel in the large image that is at the lower left corner of the small image? 5
 ii. Find the CMY coordinates of a color at (0.2, 1, 0.5) in the RGB space.
 iii. If we use direct coding of RGB values with 2 bits per primary color, how many possible colors do we have for each pixel?
 iv. What is the pitch of a color CRT?
 v. Why do many color printers use black pigment?
 [B.] Illustrate the key components of an inkjet printer and provide a brief description of the function each component performs. 4
 [C.] Describe the operational modes of 8279. 3
 [D.] Give the pinout of USB Type C 3.0. 2
- 6 [A.] Differentiate between: 5
 i. Standard and memory-mapped programmed I/O
 ii. Maskable and nonmaskable interrupts
 iii. Internal and external interrupts
 iv. Block transfer and interleaved DMA
 v. Polled and daisy chain interrupts
 [B.] Use the following chips (**Figure 6.1**) to design a microcomputer memory to include the map 3000_{16} through $33FF_{16}$. Draw a neat logic diagram. 5

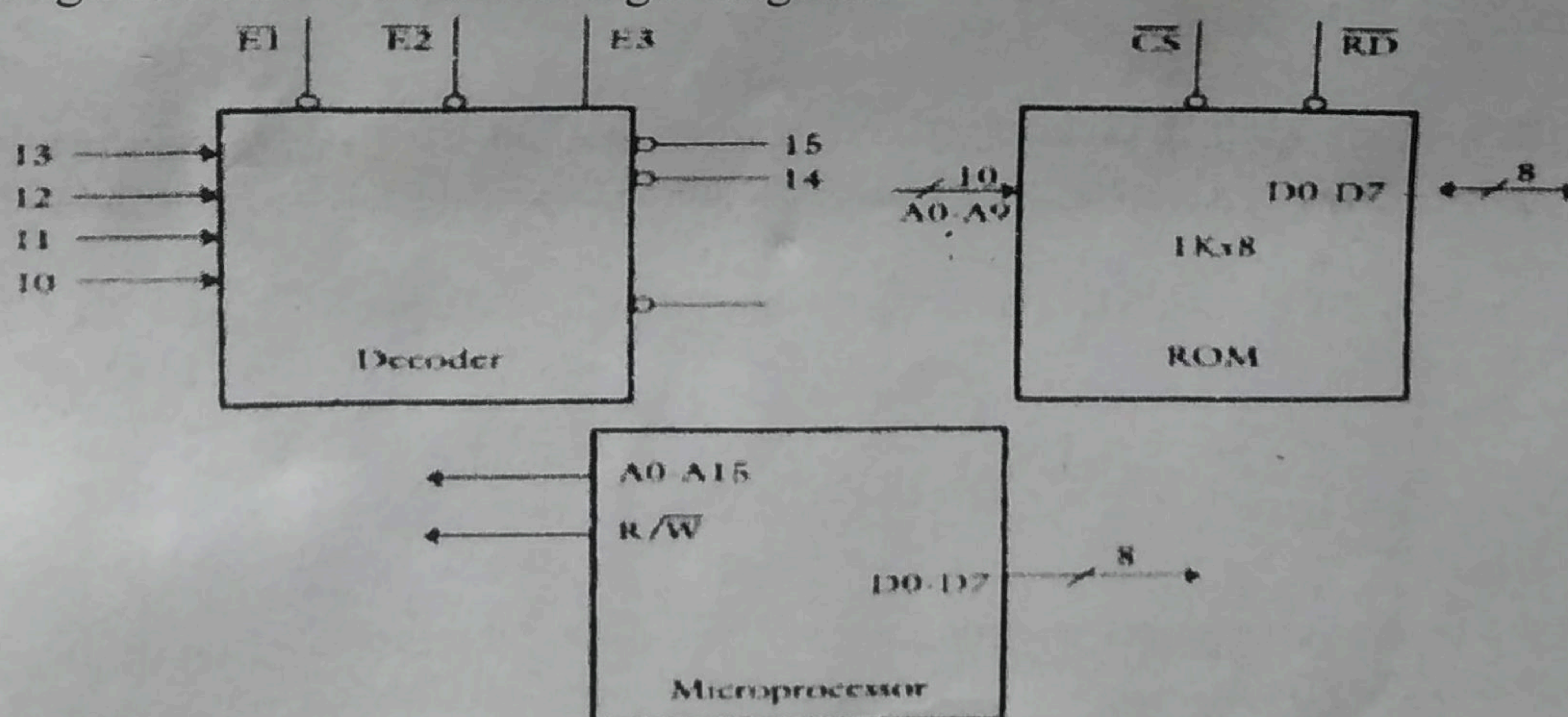


Figure 6.1

- [C.] Write short notes on
 i. Protocol Tunneling
 ii. USB4 Fabric
 [D.] What are the functions of CCD and CIS inside a scanner? X

Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

Dept. of computer Science and Information Technology

6th Semester (Level-3, Semester-II), Final Examination of B.Sc. Engg. CSE, July-December/2024

Session: 2021-22

Course Code: CIT-321 Course Title: Operating System

Full Marks: 70 Duration: 3 Hours

[Figures in the right margin indicate full marks]

Answer any 5 of the following questions.

1. (a) Explain the role of an operating system as a resource allocator and control program. How do these two views influence operating-system design? 3
- (b) Describe the differences between multiprogramming and multitasking. How does the operating system ensure reasonable response times in a multitasking environment? Critically analyze the importance of virtualization and distributed systems in modern computing environments. 4
- (c) Explain the concept of a System Call. Use a diagram or step-by-step description to show how a user application transitions from user mode to kernel mode to perform an I/O operation. 3
- (d) Compare monolithic, layered, microkernel, and modular operating-system structures. Which structure is more suitable for modern systems and why? 4
2. (a) Define a Process and describe the contents of a Process Control Block (PCB). Why is the PCB vital for context switching? 3
- (b) Distinguish between I/O-bound and CPU-bound processes. How does the degree of multiprogramming affect the balance between these two types of processes? 4
- (c) Evaluate the benefits and drawbacks of shared memory versus message passing as mechanisms for Interprocess Communication (IPC). In which scenarios is one preferred over the other? 3
- (d) Analyze the parent-child relationship in process creation. Explain the difference between the fork() and exec() system calls in a UNIX-like environment. 4
3. (a) What is the difference between a user thread and a kernel thread? Why must the operating system be aware of threads to effectively utilize multiple CPU cores? Discuss the challenges of multicore programming, specifically regarding balancing, data splitting, and data dependency. How does concurrency differ from parallelism? 4
- (b) Compare First-Come, First-Served (FCFS) and Shortest-Job-First (SJF) scheduling. Which one is prone to the convoy effect, and why? Explain the Round-Robin (RR) scheduling algorithm. Discuss how the size of the time quantum affects the performance of the system what happens if it is too large or too small? 4
- (c) Consider the following set of processes, with the length of the CPU burst given in milliseconds: 6

Process	Burst Time	Priority	Arrival
P1	2	2	2
P2	1	1	4
P3	8	4	5
P4	4	2	2
P5	5	3	0

The processes are assumed to have arrived in the order of their arrival time.

- Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, nonpreemptive priority (a larger priority number implies a higher priority), and RR (quantum = 2).
- What is the turnaround time of each process for each of the scheduling algorithms in part a?
- What is the waiting time of each process for each of these scheduling algorithms?
- Which of the algorithms results in the minimum average waiting time (over all processes)?

4. (a) A multi-process operating system must ensure proper synchronization when several processes try to access shared resources simultaneously. Identify and briefly explain the essential requirements of the Critical Section Problem that prevent conflicts and ensure fair process execution. 3

(b) In a database system shared among multiple concurrent processes, some processes only read data while others update (read/write) the database. To maintain data consistency, the Readers–Writer’s synchronization problem is solved using semaphores mutex and wrt, along with an integer variable readcount. Construct the semaphore-based solution for the Readers–Writer’s problem. Clearly write the pseudo-code for both reader and writer processes, showing how mutual exclusion is ensured, how multiple readers are allowed concurrently, and how writers obtain exclusive access to the shared database. 5

(c) A system is designed to simulate the classic *Dining Philosophers Problem* where five philosophers alternately think and eat. Each philosopher requires two chopsticks (left and right) to eat. Initially, a solution using semaphores was attempted; however, it resulted in issues such as deadlock and starvation under certain concurrent conditions. To overcome these limitations, a monitor-based solution is proposed where each philosopher can be in one of three states: THINKING, HUNGRY, or EATING, and a condition variable self[i] is used to manage waiting and signaling. 6

The key logic ensures that a philosopher i can only enter the EATING state if:

- $state[i] == HUNGRY$
- $state[(i + 4) \% 5] != EATING$
- $state[(i + 1) \% 5] != EATING$

A test(i) function is used to enforce this rule, and philosophers wait or proceed based on condition signaling.

Answer the following questions-

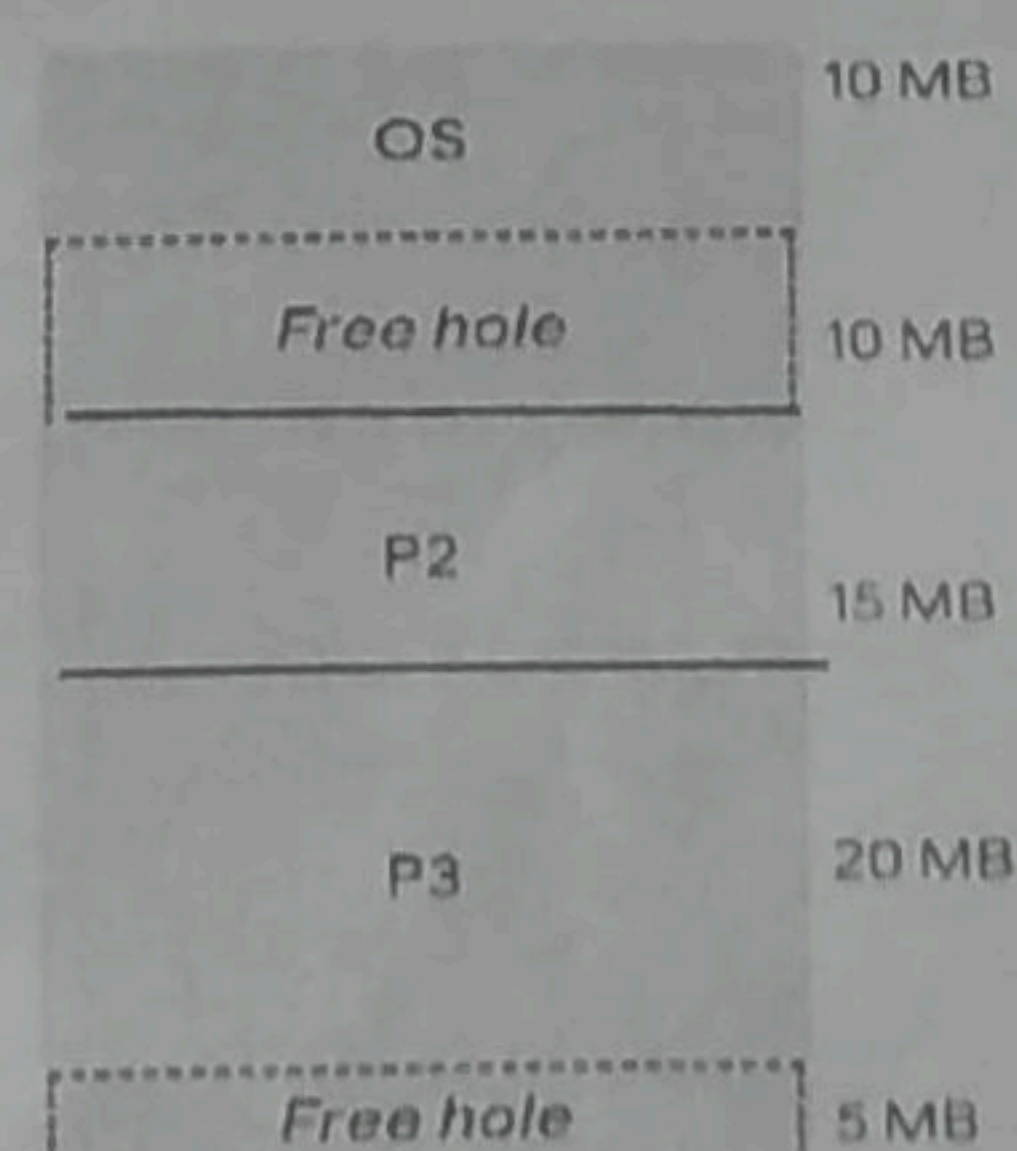
- Explain why semaphore-based solutions may fail to properly handle the Dining Philosophers Problem and justify how a monitor-based solution ensures deadlock-free synchronization.
- Write the pseudocode of the monitor-based solution including the functions pickup(i), putdown(i), and test(i).

5. (a) A system designer is analyzing a multi-threaded environment where shared resources are accessed concurrently. It is observed that the system sometimes becomes stuck, where no thread can proceed. The designer identifies that deadlock occurs only when the following four conditions hold simultaneously. Explain the necessary conditions for deadlock in a concurrent system. Also, briefly justify why all conditions must hold at the same time for a deadlock to occur. 3

(b) A computer system uses Variable Size Partitioning with Contiguous Memory Allocation in a 60 MB primary memory. The operating system occupies the first 10 MB of memory. Four processes are loaded sequentially as shown below: 5

Process	Size (MB)
(P ₁)	10
(P ₂)	15
(P ₃)	20
(P ₄)	5

Before execution



After execution

After execution, processes (P₁) and (P₄) are removed from memory. The resulting memory layout is shown in right. Now, a new process (P₅ = 12) MB arrives for allocation. Using the above information, analyze the memory allocation state and determine whether (P₅) can be allocated without compaction. Calculate the total external fragmentation, illustrate analytically how

compaction changes the memory layout, and explain why external fragmentation occurs in variable-size contiguous memory allocation systems.

- (c) A system uses the Banker's Algorithm for deadlock avoidance. The system has three resource types: $R_1 = 9$, $R_2 = 6$, $R_3 = 8$. Five processes P_0 – P_4 are competing for resources. The current system state is given below.

Process	Max Need (R_1 R_2 R_3)	Allocation (R_1 R_2 R_3)	Current Need (R_1 R_2 R_3)	Available (R_1 R_2 R_3)
P_0	5 3 4	1 0 2		
P_1	3 2 2	1 1 0		
P_2	7 4 5	2 1 1		
P_3	4 3 3	2 1 1		
P_4	6 5 4	3 2 1		

Answer the following questions-

- Write down the Safety Algorithm and Resource-Request Algorithm for process used in deadlock avoidance. Then, explain how these algorithms work together to determine whether a resource request can be safely granted without causing the system to enter an unsafe state.
- Using the Algorithm, determine whether the given system is in a safe state or an unsafe state.
- Find the safe sequence of process execution, if it exists.

6. (a) Explain the different stages of address binding in memory management and compare how these bindings differ in terms of flexibility and memory relocation support.

- (b) A modern operating system uses Hardware Support for dynamic relocation with limit and relocation registers for memory protection and address translation. Using the given memory management mechanism: Analyze the role of the limit register and relocation register in protecting process memory. Describe how the physical address is generated from a valid logical address.

Process	Limit Register	Relocation Register	Logical Address	Physical Address
P_0	380	700	210	?
P_1	425	1100	500	?
P_2	115	200	80	?

Analyze each process and determine Whether the logical address is valid or invalid. The physical address for all valid cases.

- (c) A 32-bit operating system uses a paging memory management scheme where the page size is 2 MB. A process (P_1) of size 5 MB is divided into pages and loaded into non-contiguous frames in physical memory. The page table for the process is given below:

Page Number	Frame Number
P_0	F4
P_1	F1
P_2	F6

The base addresses of the frames are:

Frame	Base Address (MB)
F1	2
F4	8
F6	12

The CPU generates the following logical addresses for the process:

- $LA_1 = 1$ MB
- $LA_2 = 4$ MB

Using the paging mechanism:

- Analyze the logical-to-physical address translation process for both logical addresses by determining the page number, offset, frame number, and physical address.
- Evaluate how the page table enables the process pages to be stored in non-contiguous memory locations.
- Identify what type of memory management problem paging helps to reduce and justify your answer based on the given scenario.

1. Answer the following:

- i. Who created Raspberry Pi?
- ii. What is the official operating system for all models of the Raspberry Pi (RPi)?
- iii. Can you netboot the Pi?
- iv. What is a GPIO pin?
- v. What will you get if you run the pinout command on Pi?
- vi. What is NOOBS?
- vii. What may happen if the Pi voltage is too low?
- viii. What connection will provide the best screen resolution?
- ix. What connection will provide the best screen resolution?
- x. Can I update Raspberry Pi firmware?
- xi. What is the RPI SoC manufacturer?
- xii. What is the 8253/54 generates a strobe in response to an externally generated signal?
- xiii. Which mode of 8253/54 generates a strobe in response to an externally generated signal?
- xiv. Which pin is used to blank the display during digit switching?
- xv. What is range of frequency for 8257?
- xvi. What is range of commonly used buffer in the serial porting?
- xvii. Which is the most commonly used buffer in the serial porting?
- xviii. What language is the Arduino IDE built on?
- xix. What does p refer to in ATmega328p?
- xx. What does p refer to in ATmega328p?
- How many times does the setup() function run on every startup of the Arduino System?
- What will happen if we supply a voltage of 25V to the Vcc of the Nokia 5110 GDM?
- What is the use of the SD.h Library in Arduino?
- Which Arduino Boards use the Atmega32U4?

2. Develop and run the Project.

per-session-mid

I. Set From A: total Marks 25

Set A-01: (Write the ratio between solved problems and total sub problems from 1 to 10)

Lab 01: A simple M/M/1 queue simulation: Write a program which performs a simple M/M/1 queue simulation. This program requires parameters for Mean Inter Arrival time of customers, Mean Service time as well as maximum number of customers. The simulation is started with a single-server queue with a FIFO queuing discipline. For M/M/1 queue, the customer inter-arrival time and the service time are both exponentially distributed. This simulation shows Average delay in queue, Average number in queue, Server utilization, and Time simulation ended.

Lab02: Inventory management: Inventory management is a crucial aspect of any business that deals with physical goods. Write a program to simulate an inventory system.

Lab 03: Inventory System Simulation (Continuous Review): Build a simulation for an inventory system where demand is normally distributed. Implement a reorder point (s) and reorder quantity (S) policy. Track total holding costs and shortage costs over a fiscal year.

Lab 04: Bernoulli data distribution: A Manufacturing and Reliability Lab is testing a newly produced batch of microchips. Each chip undergoes a strict operational test and either works perfectly (Success = 1) or fails (Failure = 0). Historical data shows that the probability of any given chip passing this test is $p = 0.85$. This lab problem models a **Bernoulli distribution** using a Quality-Control Test. It guides students through defining a binary outcome (success/failure). State the values of X and what they represent. Calculating probabilities, and finding expected metrics. Calculate the probability of a defect, $P(X=0)$. Find the mean: Calculate the expected value, $E[X]$. Find the spread: Calculate the variance, $Var(X)$.

Lab 05: Network Packet Delivery: Scenario- A network router transmits data packets in batches. Each individual packet has a success probability of $p = 0.90$ of reaching its destination. A single batch consists of $n = 15$ independent packets. Example Data: A simulation of 5 separate batches reveals the number of successfully delivered packets per batch: 14, 13, 15, 12, 14. Tasks: Identify parameters: Define the values for n , p , and q (failure probability). Exact Probability: Calculate the probability that exactly 13 packets arrive safely in a batch. Expected Value & Variance: Compute the theoretical mean μ and variance σ for the batch. Data Verification: Calculate the average number of successful packets from the example data and compare it to the theoretical mean.

Lab 06: Poisson Distribution: A customer care center receives 5 calls per hour. Compute the probability of attending zero calls, one call, two call,, ten calls per hour. Simulate the probability mass function respect to the number of receiving call. Show the probability mass function graphs considering 10 and 15 calls per hour.

Lab 07: Normal Distribution: To Show unimodal and multimodal density curves of normal distribution. To generate random sample with sample size 200, which follows a normal distribution with mean 100 and standard deviation 20. The distribution of diastolic blood pressure for men is normally distributed with a mean of about 80 and a standard deviation of 20. A histogram of the distribution of blood pressures for all men displays a normal distribution with bell shape.

Lab 08: Exponential Distribution: There were few waves in COVID-19 pandemic. Let a wave occurs every 100 days in Bangladesh, on average. After a wave occurs, find the probability using Exponential distribution that it will take more than 120 days for the next wave to occur. Simulate several Exponential distribution using rate parameters 0.5, 1.0, 2.0, and 4.0.

Simulation- Masud Sir

Lab 09: Empirical Input Modeling: Given a raw dataset of customer arrival times at a call center, fit the data to theoretical distributions (e.g., Normal, Weibull, or Lognormal) using **Maximum Likelihood Estimation (MLE)**. Validate the fit with Q-Q plots.

Lab 10: Machine Breakdown and Maintenance: Simulate a manufacturing workshop containing 5 identical machines. The time between breakdowns is exponentially distributed, and repair times follow a Weibull distribution. Calculate machine downtime and overall system throughput.

Set A-02 (Write the ratio between solved problems and total sub problems from 11 to 20)

Lab 11: Linear Congruential Generator (LCG): LCG is considered one of the basic yet best methods to generate Pseudo-Random Numbers. Write a program which generates the random number using the LCG.

Lab 12: Random variates using the inverse transform function: There are several general approaches to generating univariate Random Variable from the distribution function. Write a program which generates the desired number of random variates using the inverse transform function.

Lab 13: Generate 5,000 exponentially distributed random numbers using the Inverse Transform technique. Plot a histogram of the resulting numbers and overlay the theoretical exponential probability density function.

Lab 14: Debugging and Traceability (Verification): Write a trace program to step through a discrete-event model and log system states, ensuring it adheres exactly to the logical model rules.

Lab 15: Confidence Interval Estimation: Calculate a 95% confidence interval for the mean system response (e.g., average waiting time) by running multiple independent replications of a simulation model.

Lab 16: Steady-State Analysis: Analyze the warm-up period of a non-terminating simulation (like a continuous manufacturing line) and calculate the steady-state mean time.

Lab 17: Bank ATM System Simulation: Model an ATM vestibule where customers arrive at random, use one of several machines, and leave. Calculate the maximum queue lengths.

Lab 18: Fast-Food Drive-Thru Model: Simulate a drive-thru process (order, pay, pick up) to identify the system's operational bottlenecks.

Lab 19: Emergency Room (ER) Patient Flow: Simulate the triage, treatment, and discharge of patients in an ER to minimize waiting times and allocate resources.

Lab 20: Statistical Output Analysis and Precision Estimation for a Drive-Thru Pharmacy Simulation: This lab problem evaluates customer flow at an automated drive-thru pharmacy kiosk operating as a terminating simulation system until 50 cars are served. Students are provided with average customer waiting times across five independent replication runs (3.2, 4.3, 5.1, 4.2, and 4.6 minutes) and must apply output data analysis techniques to evaluate the system. The tasks require calculating the overall point estimate (sample mean), computing the sample variance and standard error to measure stochastic variability, and constructing a 95% confidence interval using a Student's t-distribution. Finally, students must determine the total number of simulation replications required to achieve a desired half-width precision of no more than 0.5 minutes, demonstrating how statistical confidence dictates simulation run length.

1. Set B: Marks: 25

2. Viva-Voce: 20



Patuakhali Science and Technology University (PSTU)

Faculty of Computer Science and Engineering

B.Sc. Engg. 3rd Year 2nd Semester, Final Examination, January-June 2025

Course Title: Operating System Sessional
Time: 01 Hour

Course Code: CIT 322

Full Marks: 20

N.B.: Figures shown in the right margin indicate full marks

- | | | |
|--------|--|--------------|
| | | Marks |
| 1. (a) | <p>A university design laboratory contains 5 designers working on different projects. There are 5 shared 3D printers arranged in a circular layout. Each designer requires two adjacent printers (the printer on the left and the printer on the right) to complete a printing job because their models are printed in two synchronized parts.</p> <p>A designer repeatedly performs the following activities:</p> <ul style="list-style-type: none">• Designing a model• Waiting for printers• Printing the model <p>A designer may begin printing only if both adjacent printers are available. After finishing the print job, the designer releases both printers so neighboring designers may use them. To avoid resource conflicts, the laboratory management requires that the synchronization mechanism must prevent deadlock while ensuring that no two neighboring designers use the same printer simultaneously.</p> <p>Implement the above synchronization problem in C++ using Monitors. Your program must satisfy the following requirements:</p> <ol style="list-style-type: none">i. There are 5 designers and 5 shared printers.ii. Each designer repeatedly performs:<ol style="list-style-type: none">a. Designingb. Waiting for printersc. Printingiii. A designer can start printing only when both adjacent printers are available.iv. Use a Monitor (implemented with mutex and condition_variable) for synchronization.v. Ensure that:<ol style="list-style-type: none">a. No two neighboring designers print simultaneously.b. Deadlock does not occur.c. Shared printers are released immediately after printing.vi. Display the state transitions of each designer, such as:<ol style="list-style-type: none">a. Designingb. Waiting for Printersc. Printingvii. Simulate the behavior using multiple threads, where each designer performs three printing cycles. | 10 |

Sample Output

Designer 1 is Designing.
Designer 2 is Designing.
Designer 3 is Designing.
Designer 4 is Designing.
Designer 5 is Designing.

Designer 2 is Waiting for Printers.
Designer 2 starts Printing.

Designer 4 is Waiting for Printers.
Designer 4 starts Printing.

Designer 2 finishes Printing and releases both printers.

Designer 1 starts Printing.

- (b) A university data center manages multiple processes that compete for limited hardware resources. To ensure that deadlock never occurs, the operating system uses the Banker's Algorithm before granting any resource request.

The system contains three resource types:

- R1 = 10 instances
- R2 = 5 instances
- R3 = 7 instances

Five processes (P0–P4) are currently executing. Their Maximum Demand and Current Allocation are shown below.

Process	Max Need (R1 R2 R3)	Allocated Resource (R1 R2 R3)	Current Resource Need (R1 R2 R3)	Available Resource (R1 R2 R3)
P0	7 5 3	0 1 0		
P1	3 2 2	2 0 0		
P2	9 0 2	3 0 2		
P3	2 2 2	2 1 1		
P4	4 3 3	0 0 2		

The operating system must determine whether the current allocation is safe before allowing further execution.

Write a C++ program that performs the following operations.

1. Store the Maximum and Allocation matrices.
2. Calculate the Need Matrix using

$$\text{Need} = \text{Maximum} - \text{Allocation}$$
3. Calculate the Available vector from the total resources.
4. Implement the Safety Algorithm to determine whether the current system is:
 - Safe
 - Unsafe
5. If the system is safe, print
 - Available vector
 - Need matrix
 - Safe sequence of process execution
6. Otherwise print - **System is in an Unsafe State.**

Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

Dept. of Computer and Communication Engineering

Final Examination of B. Sc. Engineering in CSE Level: 3 Semester: II Session: 2021-2022

Course Code	Course Title	July December 2024	Credit: 1.50
CCE-322	Computer Peripheral and Interfacing Sessional		Marks: 70

1. Draw the circuit diagram of your design. Indicate the components. 5
2. Illustrate the cost-benefit analysis for your project. 5
3. Make the project report. 10
4. Develop and display the Project. 30
5. Attend viva-voce. 20